

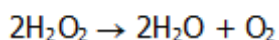
Unit 5 MC Practice

Use the following information to answer questions 1–3.

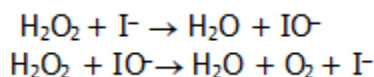
The following experimental data was collected for the hypothetical decomposition reaction $A \rightarrow B + C$.

Experiment	[A] (M)	T (K)	$t_{1/2}$ (min)
1	0.200	298	0.270
2	0.400	298	0.270
3	0.200	273	0.350
4	0.400	273	0.350

- Based on the information above, what is the order of the reaction?
 - The reaction is first order, because the stoichiometric coefficient of the reactant is 1.
 - The reaction is first order, because half-life is independent of reactant concentration.
 - The reaction is zero order, because changing the reactant concentration did not affect the half-life.
 - The reaction is zero order, because decreasing the temperature of the reaction increased the half-life.
- For which experiment is the initial rate of the reaction highest?
 - Experiment 1, because fewer reactant molecules will need to be converted into product.
 - Experiment 2, because both the temperature and reactant concentration are maximized.
 - Experiment 3, because a lower temperature will decompose A most efficiently.
 - Experiment 4, because a lower temperature and higher [reactant] maximizes reaction rate.
- Calculate the rate constant, k , for this reaction at 273 K.
 - 2.86 min^{-1}
 - $2.86 \text{ min}^{-1} \text{ M}^{-1}$
 - 1.98 min^{-1}
 - $1.98 \text{ min}^{-1} \text{ M}^{-1}$



4. The decomposition of hydrogen peroxide is shown above and has the following proposed mechanism:



Which of the following substances serves as a catalyst for this reaction?

- H_2O_2
 - I^-
 - H_2O
 - IO^-
5. For the reaction whose rate law is given below, a plot of which of the following is a straight line?
- $$\text{rate} = k[\text{X}]$$
- $[\text{X}]$ vs time
 - $1/[\text{X}]$ vs time
 - $\ln[\text{X}]$ vs 1/time
 - $\ln[\text{X}]$ vs time

6. $A + B \rightarrow C + D$ $\text{rate} = k[\text{A}][\text{B}]^2$

What are the potential units for the rate constant for the above reaction?

- s^{-1}
- $\text{s}^{-1} \text{ M}^{-1}$
- $\text{s}^{-1} \text{ M}^{-2}$
- $\text{s}^{-1} \text{ M}^{-3}$

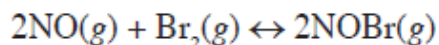
Use the following information to answer questions 7–8.

Reactant A underwent a decomposition reaction: $A \rightarrow 2B + C$. The concentration of A was measured periodically and recorded in the chart below.

Time (Hours)	[A] <i>M</i>
0	0.40
1	0.20
2	0.10
3	0.05

7. Based on the data in the chart, which of the following is the rate law for the reaction?
- a. $\text{rate} = k[A]$ b. $\text{rate} = k[A]^2$ c. $\text{rate} = 2k[A]$ d. $\text{rate} = \frac{1}{2} k[A]$
8. What would be the concentration of B after two hours?
- a. 0.15 M b. 0.20 M c. 0.30 M d. 0.60 M

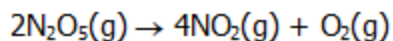
Use the following information to answer questions 9–10.



The above experiment was performed several times, and the following data was gathered:

Trial	$[\text{NO}]_{\text{init}}$ (<i>M</i>)	$[\text{Br}_2]_{\text{init}}$ (<i>M</i>)	Initial Rate of Reaction (<i>M/min</i>)
1	0.20 <i>M</i>	0.10 <i>M</i>	5.20×10^{-3}
2	0.20 <i>M</i>	0.20 <i>M</i>	1.04×10^{-2}
3	0.40 <i>M</i>	0.10 <i>M</i>	2.08×10^{-2}

9. What is the rate law for this reaction?
- a. $\text{rate} = k[\text{NO}][\text{Br}_2]^2$ b. $\text{rate} = k[\text{NO}]^2[\text{Br}_2]^2$ c. $\text{rate} = k[\text{NO}][\text{Br}_2]$ d. $\text{rate} = k[\text{NO}]^2[\text{Br}_2]$
10. What would be the initial rate of the reaction if $[\text{NO}]_{\text{init}} = 0.40 \text{ M}$ and $[\text{Br}_2]_{\text{init}} = 0.20 \text{ M}$?
- a. $2.08 \times 10^{-2} \text{ M/min}$ b. $4.16 \times 10^{-2} \text{ M/min}$ c. $6.24 \times 10^{-2} \text{ M/min}$ d. $8.32 \times 10^{-2} \text{ M/min}$
11. A proposed mechanism for a reaction is as follows:
- $\text{NO}_2 + \text{F}_2 \rightarrow \text{NO}_2\text{F} + \text{F}$ slow step
- $\text{F} + \text{NO}_2 \rightarrow \text{NO}_2\text{F}$ fast step
- What is the order of the overall reaction?
- a. zero order b. first order c. second order d. third order



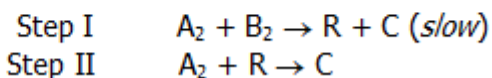
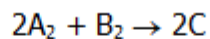
12. The following data was collected at 25°C and 1 atm of pressure for the reaction shown above.

Time (minutes)	[N ₂ O ₅] (mol/L)
0	1.24×10^{-2}
10.	0.92×10^{-2}
20.	0.68×10^{-2}
30.	0.50×10^{-2}
40.	0.37×10^{-2}
50.	0.28×10^{-2}
70.	0.15×10^{-2}

Which of the following best represents the half-life for this reaction?

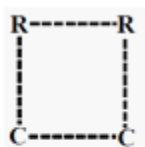
- a. 15 min b. 18 min c. 23 min d. 36 min

Use the following information to answer questions 13–14.



13. The activated complex that forms in Step I of the mechanism shown above could have which of the following structures? (The dotted lines represent partial bonds.)

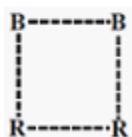
A



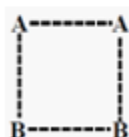
C



B

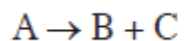


D



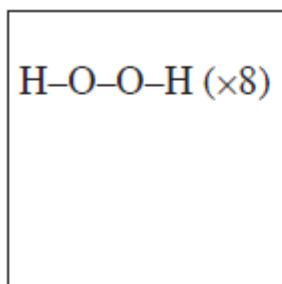
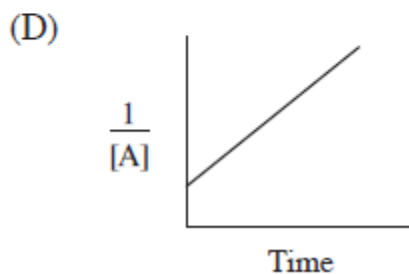
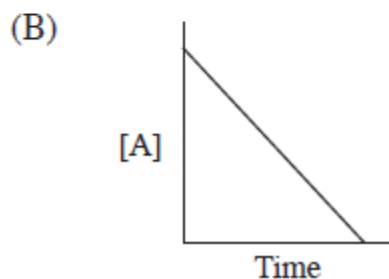
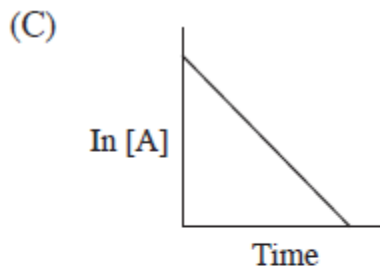
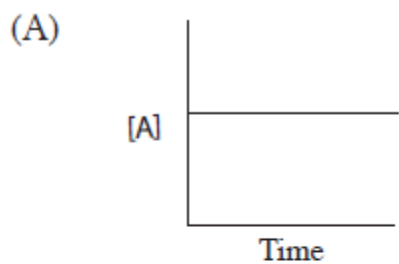
14. Which of the following rate law expressions best agrees with the proposed mechanism?

- a. rate = $k[\text{A}_2]$ b. rate = $k[\text{A}_2]^2$ c. rate = $k[\text{A}_2][\text{R}]$ d. rate = $k[\text{A}_2][\text{B}_2]$

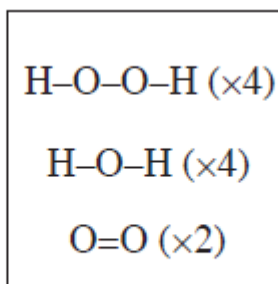


$$\text{rate} = k[A]^2$$

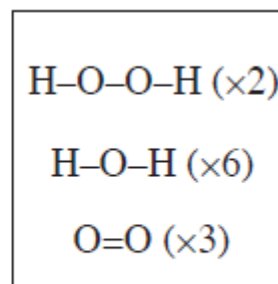
15. Which of the following graphs may have been created using data gathered from the above reaction?



$t = 0\text{s}$



$t = 200\text{s}$



$t = 400\text{s}$

16. The above diagrams show the decomposition of hydrogen peroxide in a sealed container in the presence of a catalyst. What is the overall order for the reaction?

- a. zero order b. first order c. second order d. third order